# Network Operations & Predictive Intelligence System

From reactive network monitoring to proactive network intelligence

# The Problem

**Telecom networks generate massive amounts of activity data**

Every hour, thousands of network cells generate data from:

- SMS activity
- Call activity
- Internet usage
- Geographic grid locations
- Historical network behaviour

# The challenge

Traditional network monitoring is largely reactive.

A network operations team often discovers an abnormal situation only after activity has already changed significantly.

This creates problems:

- Delayed detection
- Manual investigation
- Difficulty identifying high-risk cells
- Large volumes of data to monitor
- Historical patterns are difficult to use operationally

# Key question:

Can we predict which network cells are likely to experience unusually high activity before the NOC needs to investigate them?

# The Product

## Network Operations Intelligence Platform

A unified platform that transforms raw telecom activity data into:

**Observe**

Real-time network activity visibility

**Detect**

Historical anomaly detection

**Predict**

Machine-learning-based network risk prediction

**Explain**

AI-assisted operational insights

**Act**

Give the NOC actionable information instead of raw data

# From Raw Data to Intelligence

Telecom Activity Data ->

Data Ingestion ->

Data Cleaning & Standardization ->

Spark Processing ->

MySQL Analytical Warehouse ->

ML Features ->

ML Prediction & Anomaly Detection ->

API Layer ->

React Dashboard ->

Claude AI Assistant

| Question | Component |
|---|---|
| **What happened?** | Spark / Analytics |
| **Where?** | GeoJSON + Map |
| **Is it unusual?** | Alerts / Anomaly Detection |
| **How risky is it?** | ML |
| **Is our data trustworthy?** | Pipeline Status |
| **What should I investigate?** | Claude |
| **Why does Claude believe that?** | Tool calls + Evidence |

# Six behavioural features

**Average Activity**

How active has the grid been recently?

**Activity Growth**

Is activity increasing compared with the previous period?

**Active Hours**

How consistently active is the grid?

**Peak Ratio**

How strong are peaks compared with the average?

**Variability**

How much does activity fluctuate?

**Internet Share**

How much of the activity comes from internet usage?

# Plus time-of-day features

`hour_sin`

`hour_cos`

# NOC Dashboard

## One Operational View

The React dashboard brings the different intelligence layers together.

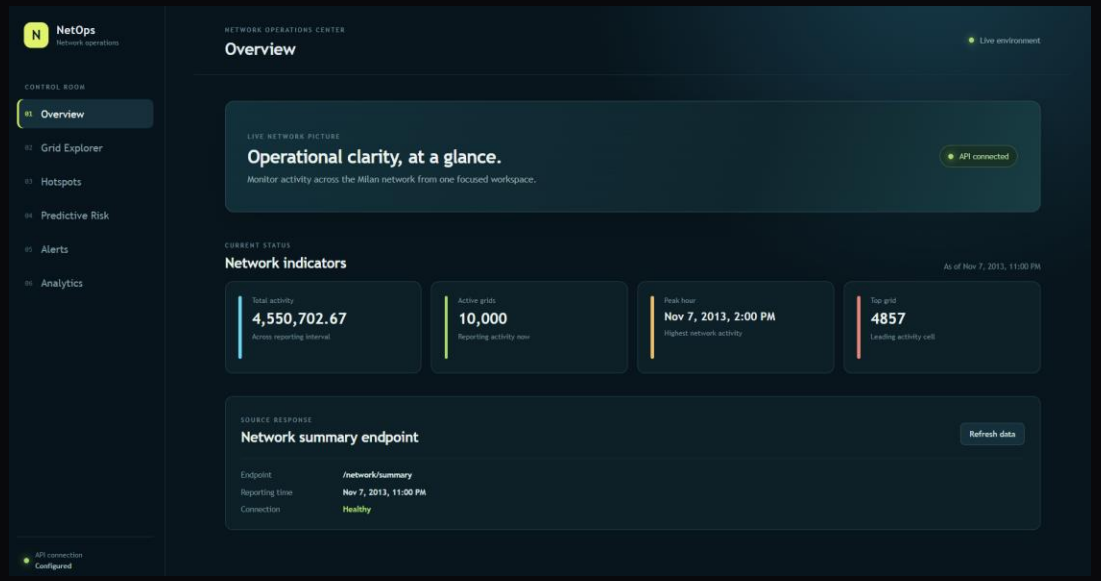
**Dashboard capabilities:**

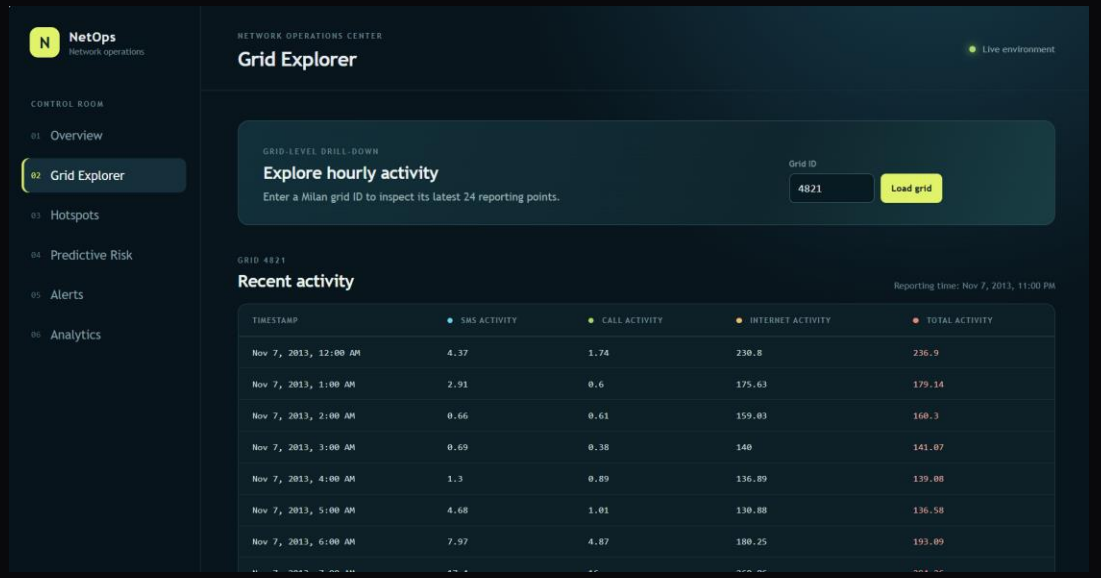
## Network Overview

- Total activity
- Active grids
- Network statistics

## Grid Monitoring

- Individual grid activity
- Historical behaviour



## Hotspots

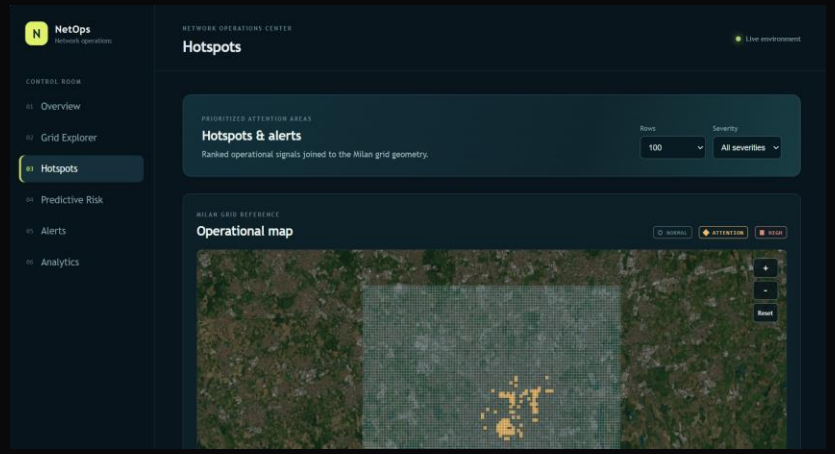
- Highest activity grids

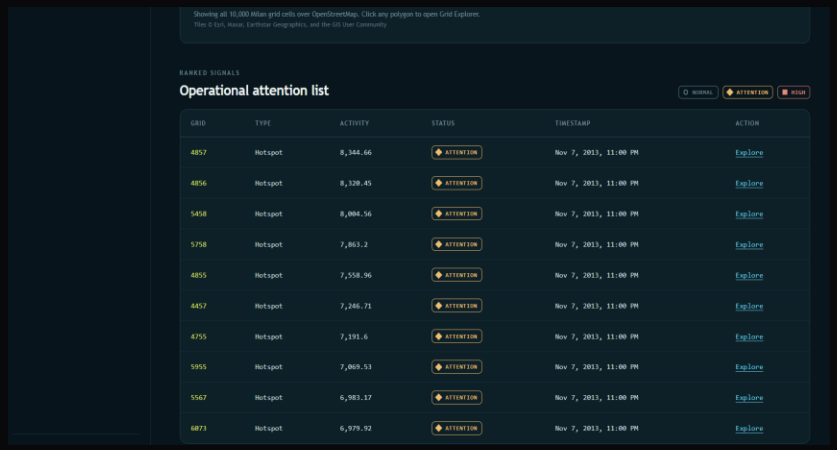
## Alerts

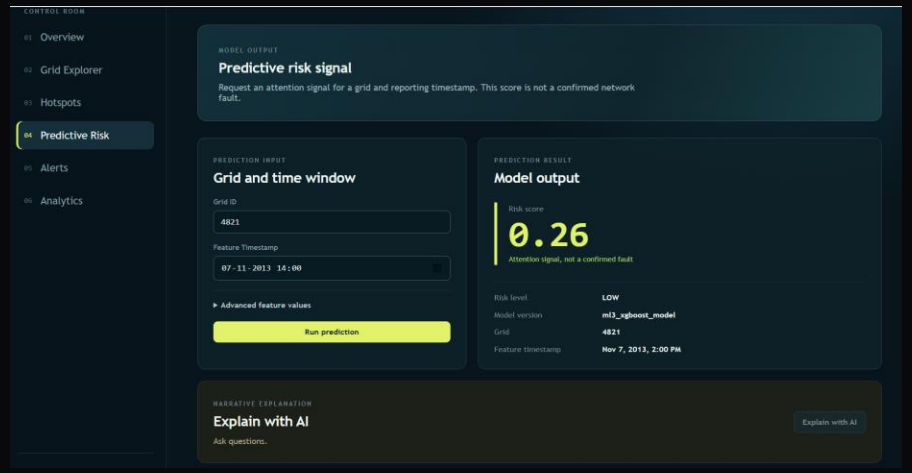
- Network alerts
- Historical anomalies

## Predictive Risk

- Select grid
- Select timestamp
- Receive ML prediction



# AI Assistant

## Claude-Powered Network Assistant

Instead of forcing an operator to manually interpret multiple screens and datasets, the AI assistant provides a natural-language interface.

**Example questions**

"Why is Grid 4821 high risk?"

"What happened to this grid historically?"

"Which grids should I investigate?"

"Is this activity unusual for this time of day?"

# Value to Network Operations

**Faster detection**

Quickly surface unusual areas.

**Better prioritization**

Focus operators on the grids that need attention.

**Geographic awareness**

Connect analytical results to physical locations.

**Evidence-based investigation**

Claude works from actual platform evidence.

**Operational transparency**

Pipeline health and uncertainty are visible instead of hidden.